Structural Self-Energy Expansion (SSEE)

A minimal-parameter dark energy framework from φ and π

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Seeking arXiv endorsement in astro-ph.CO (or gr-qc)

Core idea in one sentence

The core cosmological background — w_0, w_a, H_0, Ω_m, n_s — is derived *algebraically* from the golden ratio $\varphi = (1+\sqrt{5})/2$ and π with **zero fitted dimensionless parameters at the background level**. Dark-matter sector observables (α_K, m_φ) are formulated as a working EFT proposal subject to full Boltzmann validation, with a handful of phenomenological ansätze (m_φ OP-9, ξ OP-11) tracked as open problems in `OPEN_PROBLEMS.md` (OP-9/11/14; the former MIRA matter mechanism OP-8 is now dissolved under the ω_m-direct reframe).

Key algebraic predictions

Quantity	SSEE algebraic value	Observed / constraint	Agreement
w_0	$-T_r/M_v = -0.8400$	DESI DR2: -0.838 ± 0.055	0.24σ
w_a	$-P_{sc}/I_g = -0.669975$	DESI DR2: -0.631 ± 0.226	0.17σ
$H_0^{\rm alg}$	$3(\varphi+\pi)^2 = 67.962$ km/s/Mpc	Planck 2018: 67.36 ± 0.54	1.11σ
$\Omega_{m,\rm CMB}$	$\omega_m/h^2 = 0.308881$ (ω_m-direct)	Planck 2018: 0.3153 ± 0.0073	0.88σ
r_d	CAMB (ω_m-direct): 147.17 Mpc	Planck 2018: 147.09 ± 0.26 Mpc	0.32σ
$H_0^{\rm local}$	$H_0^{\rm alg}/(1-f_{\rm scr}) = 72.86$ km/s/Mpc	SH0ES 2022: 73.04 ± 1.04	0.17σ
n_s	$1-\varphi^{-7} = 0.965558$	Planck PR4: 0.9649 ± 0.0042	0.16σ
r	$\varphi^{-10} = 0.008131$	BK18: $r < 0.036$	Prediction
$\alpha_K(0)$	$3\Omega_{\rm DE}(1+w_\phi) = 0.072567$	Euclid forecast sensitivity	< 0.1
m_φ	$\Sigma m_\nu^{\rm active}(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V) = 40.70$ eV	Matter $P(k)$ via $k_{\rm fs}$	Prediction
$k_{\rm fs}$	$0.754\ h$/Mpc	DESI Y3/Euclid (2026–28)	Prediction

Ten-paper series (all reproducible)

I. Derived Algebraic Background

- **Paper 1 (Framework)**: Algebraic derivation of geometric constants; α-attractor embedding ($\alpha = \varphi^4/3$). [25 pp]
- **Paper 4 (Nine Sovereignties)**: CMB observables (ℓ_1, ℓ_2, ℓ_3, θ_*, A_s, n_s, τ, r) bounded algebraically. [17 pp]

II. Tested Against Data

- **Paper 2 (MCMC)**: 100 walkers $\times$ 25 000 steps vs. DESI DR2 + Planck + 7 clusters; $\Delta\mathrm{BIC} = -6.43$ (SSEE favoured, $k_{\rm SSEE} = 2$ vs. $k_{\Lambda\rm CDM} = 3$), total matter density in the background geometry. [26 pp]
- **Paper 3 (CMB Confrontation)**: CAMB ω_m-direct ($\Omega_m = 0.308881$); Cobaya full `plik` MCMC; $\Delta\mathrm{BIC} = -32.9$ (canonical; -24.0 in the `plik_lite` point estimate), SSEE favoured under $k=2$ vs $k=6$ model comparison. [24 pp]

III. Phenomenological Extensions & Open Directions

- **Paper 5 (Israel–Stewart Perturbations)**: Causal bulk viscosity; $c_s^2 = 0$ exact; $\gamma_{\rm IS} = 0.5504 \approx \gamma_{\Lambda\rm CDM}$. [25 pp]

- **Paper 6 (φ-DM Two-Sector)**: $m_\varphi = 40.70$ eV from the algebraic mass relation $m_\varphi = \Sigma m_\nu^{\text{active}}(\text{SOLAR}^2 \cdot \text{KRYSTOS}_V)$; proposes a two-sector model that reconciles the S_8 lensing tension ($S_8^{\text{eff}} = 0.758$, 0.04σ KiDS) with an $f\sigma_8$ tension of 0.93σ ($< 1\sigma$; single-sector baseline 0.70σ) at the linear-growth level. [24 pp]
- **Paper 7 (Canonical EFT Proposal)**: $\beta_c = -\text{AURA}$ (numerical -3.998); $\alpha_K = 0.072567$ algebraic; fundamental parameters λ, V_0, M proposed. [16 pp]

IV. Strong-Field Regime, Hubble Tension & UV Completion

- **Paper 8 (Strong Gravity)** [*draft, pending OP-9*]: Disformal null geodesic analysed in two limits: alternative MOND-like (deflection $\sqrt{\text{AURA}} \approx 1.9995$, numerically close to $\mathcal{M} = \text{AURA}/2$ at 0.03% but algebraically distinct) and canonical two-sector with EFT $\alpha_B = \alpha_M = \alpha_T = 0$ (residual fifth-force $\sim 10^{-15}$ at kpc, observable lensing recovers GR); selective coupling satisfies solar-system tests. Current SLACS/BELLS data favour the canonical limit (OPEN_PROBLEMS.md OP-13). [17 pp]
- **Paper 9 (Hubble Tension)**: Algebraic local screening fraction $f_{\text{scr}} = (\pi - \varphi)/(\varphi + \pi)^2 \approx 0.067$ applied to the single global anchor $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.962$ gives the canonical $H_0^{\text{local}} = 72.86$ km/s/Mpc, 0.17σ from SH0ES, an algebraic prediction with no fitted parameters (2026 reframe; the earlier CMB-mapped $H_0^{\text{MIRA}} = 67.04 \to 71.87$, 1.12σ, is superseded). [18 pp]
- **Paper 10 (UV Completion)**: Higher-derivative operator $K(X) = X/\text{KAL}_0 + X^2/M^4$ with algebraic cutoff $M = 9.68$ meV; connects the dark-energy EFT to the inflationary α-attractor. [14 pp]

Falsifiability (explicit)

- $k_{\text{fs}} = 0.754\ h$/Mpc — set by the algebraic mass $m_\varphi = 40.70$ eV; since φ-DM has no Standard-Model portal, the falsifiable channel is the matter power spectrum (DESI Y3 / Euclid), not neutrino-mass experiments.
- $\alpha_K \approx 0.073$ — Euclid weak lensing will probe this kineticity limit.
- $r = \varphi^{-10} = 0.008131$ — LiteBIRD (~2032) will confirm or refute the tensor-to-scalar ratio.
- A $> 3\sigma$ deviation of w_0 from -0.840 would falsify the corresponding background sector of the framework.

What I am asking for

An arXiv endorsement in `astro-ph.CO` (or `gr-qc`) to submit Papers 1–3 as the first tranche. All code, data, and LaTeX sources are public at github.com/mikealmeida1721/SSEE under Apache-2.0.

I am happy to answer any technical questions or share the PDFs directly. Thank you for considering this request.

Code: Python (emcee, CAMB, CLASS), all reproducible in <30 min. **Data:** DESI DR2 (public), Planck PR4 (public), KiDS-1000 (public).